LESSON 6.11

Fraction Division Problem Solving

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve real-world problems by deciding whether the question asks how many groups fit or how much is in one group, writing a fraction division equation, and checking the solution by multiplying.

WHY THIS MATTERS Agent Torres is closing the biggest case of the year. She has 2/3 of a gallon of fingerprint dust left and each test uses 1/6 of a gallon. Before the evidence expires, she needs to figure out exactly how many tests she can run. Can you write the equation and solve it?

REMEMBER

Fraction Division Problem Solving

Fraction Division Word Problems. Formula: Total Amount ÷ Group Size = Number of Groups In a word problem, first decide what the question asks: how many groups fit (total ÷ size of each group) or how much is in one group (total ÷ number of groups). The total amount is the dividend and the other number is the divisor. Write the equation, solve it with a model, common denominators, or by multiplying by the reciprocal, then check by multiplying the quotient by the divisor.

1. Identify the Dividend (total amount being split or shared).
2. Identify the Divisor (size of each group or part).
3. Divide: Total ÷ Size = Number of Groups (or Total ÷ Groups = Size per Group)

Words: **Model *(Modelo)*** **Equation *(Ecuación)*** **Solution *(Solución)*** **Reasonableness *(Razonabilidad)*** **Inverse operations *(Operaciones inversas)***

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

A forensic chemist has 5/6 of a liter of testing solution. Each DNA test requires 1/12 of a liter. She needs to figure out how many tests she can run before the solution runs out. Which equation matches the problem?

(A) 1/12 ÷ 5/6 = t
(B) 5/6 ÷ 1/12 = t
(C) 5/6 × 1/12 = t
(D) 5/6 + 1/12 = t

SHOW YOUR WORK

PART 2 Practice the skill

show every step

2 Circle the letter of the best answer. Show how you know.

What is 2 1/2 ÷ 1/2?

(A) 1 1/4
(B) 2 1/4
(C) 4
(D) 5

SHOW YOUR WORK

3 Circle the letter of the best answer. Show how you know.

What is 1 1/2 ÷ 3/4?

(A) 1 1/8
(B) 2
(C) 3/4
(D) 9/8

SHOW YOUR WORK

4 Circle the letter of the best answer. Show how you know.

What is 3 1/2 ÷ 1/4?

(A) 3 1/8
(B) 7/8
(C) 13/4
(D) 14

SHOW YOUR WORK

PART 3 Apply it

real-world problems

5 **Circle the letter of the best answer. Show how you know.**

Three agents share 3/4 pound of casting clay equally. How much clay does each agent get? (3/4 ÷ 3 asks how much is in one share.)

(A) 2 1/4 pounds

(B) 4 pounds

(C) 3 3/4 pounds

(D) 1/4 pound

HINT Is each share more or less than the 3/4 pound you started with?

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Convert 3 2/5 to an improper fraction.

(A) 6/5

(B) 15/5

(C) 17/5

(D) 32/5

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

EXPLAIN YOUR THINKING

How do you decide which value is the dividend and which is the divisor in a word problem?

SENTENCE STARTER The total amount is the ___ because it is being split into groups. The size of each group is the ___ because we are dividing by that amount.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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LEARNING TARGET I can solve real-world problems by deciding whether the question asks how many groups fit or how much is in one group, writing a fraction division equation, and checking the solution by multiplying.

WHY THIS MATTERS Agent Torres is closing the biggest case of the year. She has 2/3 of a gallon of fingerprint dust left and each test uses 1/6 of a gallon. Before the evidence expires, she needs to figure out exactly how many tests she can run. Can you write the equation and solve it?

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Room 3 — Sort the Clues: Each safe opens only if you pick the clues that need fraction DIVISION. Sort each clue into the correct lock.

Groups: Needs Division (how many groups fit?) | Needs Multiplication (a part of an amount)

_____ A roll of crime-scene tape is 3/5 mile long, cut into 1/10-mile strips. How many strips? (3/5 ÷ 1/10)

_____ A footprint trail is 2/3 mile long with markers every 1/12 mile. How many markers? (2/3 ÷ 1/12)

_____ Each evidence bag weighs 1/4 pound. The detective grabs 4 bags. What is the total weight? (1/4 × 4)

_____ A report is 3/8 page long. Torres reads half of it. How much did she read? (3/8 × 1/2)

2 **Write the letter of the matching item on each line.**

Room 4 — Match the Lock: Drag each word problem to the division equation that unlocks it.

_____ A 4/5-yard cord is cut into 1/10-yard fuses for the alarm.

_____ A 1/2-gallon jug of luminol is poured into 1/8-gallon sprayers.

_____ A 3/4-mile escape route is split into 3/8-mile sprints.

_____ A 5/6-liter sample is divided into 1/12-liter test tubes.

(A) 4/5 ÷ 1/10 = 8

(B) 1/2 ÷ 1/8 = 4

(C) 3/4 ÷ 3/8 = 2

(D) 5/6 ÷ 1/12 = 10

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A forensic chemist has 5/6 of a liter of testing solution. Each DNA test requires 1/12 of a liter. She needs to figure out how many tests she can run before the solution runs out. 5/6 ÷ 1/12 = ?

(A) 2

(B) 5/72

(C) 10

(D) 12

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A forensic chemist has 5/6 of a liter of testing solution. Each DNA test requires 1/12 of a liter. She needs to figure out how many tests she can run before the solution runs out. Is 10 tests a reasonable answer?

(A) No — the answer should be less than 1

(B) Yes — each test uses far less than the total, so the answer should be well above 1

(C) No — you cannot divide fractions

(D) Yes — because 10 is a round number

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Room 1 — Evidence Locker: Agent Torres has 7/8 pound of casting clay to lift footprints. Each cast uses 1/8 pound. How many casts can she make to unlock the door?

(A) 1/7

(B) 6

(C) 7

(D) 8

Rewrite each fraction _____

Work it out _____

Simplify _____

6 **Circle the letter of the best answer. Show how you know.**

Room 2 — Surveillance Map: A suspect's trail is 5/6 of a mile. Hidden checkpoints are spaced every 1/6 of a mile. How many checkpoints are on the trail?

(A) 1/5

(B) 4

(C) 5

(D) 6

Rewrite each fraction _____

Work it out _____

Simplify _____

EXPLAIN YOUR THINKING

How do you decide which value is the dividend and which is the divisor in a word problem?

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